

# MACHINE LEARNING INTERNSHIP

## TASK 3: TITANIC SURVIVAL PREDICTION

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Colab Notebook: [Task 3 Internspark ML.ipynb](#)

### 1. PROJECT OVERVIEW

The objective of this project is to build a machine learning classification model capable of predicting whether a passenger survived the Titanic disaster based on passenger attributes. The project emphasizes feature engineering, handling missing values, model training, evaluation, and model explainability.

### 2. DATASET INFORMATION

Dataset Name: Titanic Survival Prediction Dataset

Target Variable:

- Survived

Important Features:

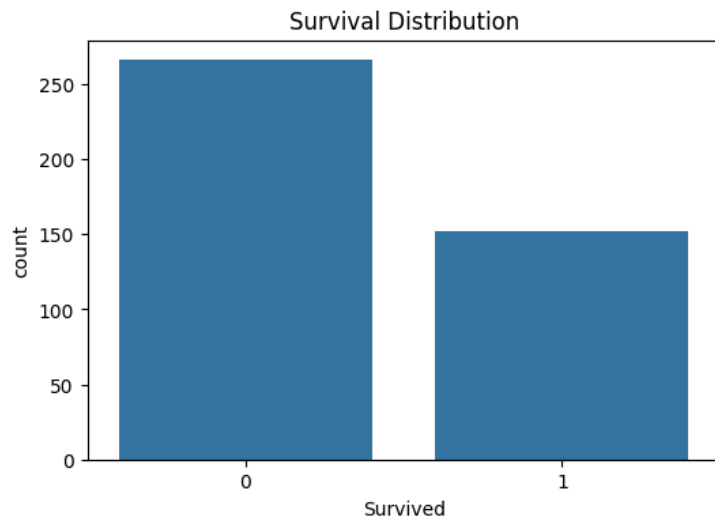
- Passenger Class (Pclass)
- Sex
- Age
- Fare
- Embarked
- SibSp
- Parch
- Cabin Information

Feature Engineering Performed:

- Title Extraction from passenger names
- Family Size creation
- Cabin Presence indicator

### 3. EXPLORATORY DATA ANALYSIS

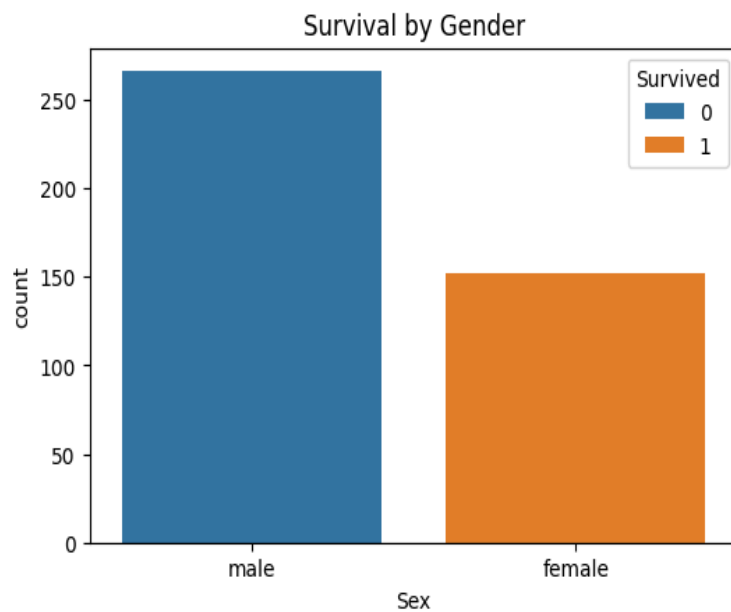
Figure 1: Survival Distribution



Observation:

The dataset contains more passengers who did not survive than those who survived. Approximately 60% of passengers belong to the non-survivor class, while the remaining passengers survived. This indicates a slightly imbalanced dataset.

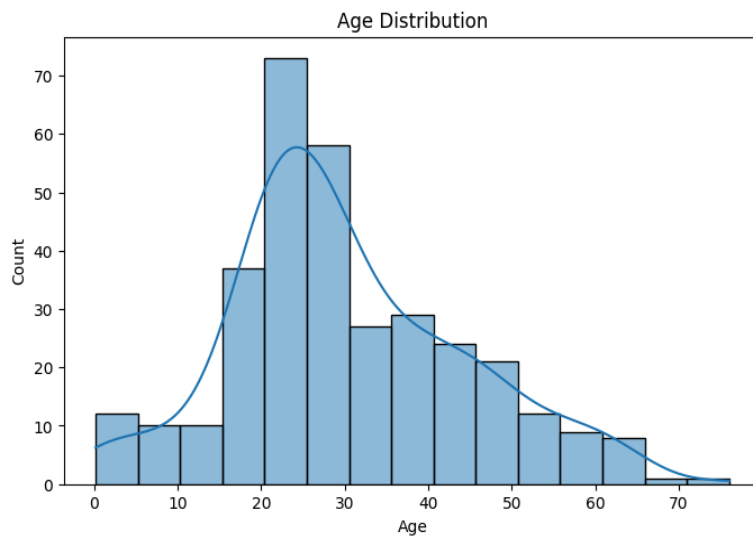
Figure 2: Survival by Gender



Observation:

Gender has a strong impact on survival. Female passengers show a significantly higher survival rate compared to male passengers. This suggests that gender is one of the most important predictors of survival.

Figure 3: Age Distribution



Observation:

Most passengers fall within the age range of approximately 20 to 35 years. The distribution is slightly right-skewed, with fewer elderly passengers. Missing age values were handled using median imputation during preprocessing.

#### 4. DATA PREPROCESSING

The following preprocessing steps were performed:

- Missing values in Age were filled using the median value.
- Missing values in Fare were filled using the median value.
- Cabin information was converted into a Cabin\_Present feature.
- Categorical variables were encoded using Label Encoding.
- Numerical features were standardized using StandardScaler.

#### 5. FEATURE ENGINEERING

To improve model performance, several new features were created:

Title Extraction:

Passenger titles such as Mr, Mrs, Miss, and other honorifics were extracted from names.

Family Size:

Family size was calculated using the formula:

$$\text{Family Size} = \text{SibSp} + \text{Parch} + 1$$

Cabin Presence:

A binary feature was created indicating whether cabin information was available for a passenger.

These engineered features helped capture additional information not directly available in the original dataset.

## 6. MODEL TRAINING

The following classification algorithms were trained and evaluated:

1. Logistic Regression
2. Random Forest Classifier

The dataset was divided into training and testing subsets using an 80:20 split ratio.

## 7. MODEL EVALUATION

Evaluation Metrics Used:

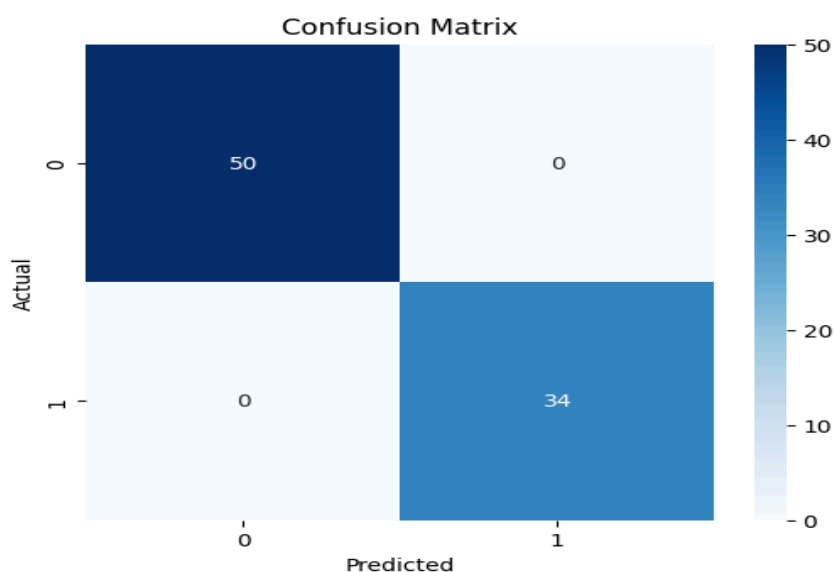
- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-Score

Observation:

Both models achieved strong performance. Random Forest was selected as the final model due to its superior ability to capture complex relationships between passenger attributes and survival outcomes.

## 8. CONFUSION MATRIX

Figure 4: Confusion Matrix



Observation:

The confusion matrix shows perfect classification results on the testing dataset. All 50 non-survivors were correctly classified and all 34 survivors were correctly identified. No misclassifications occurred.

Results:

True Negatives = 50

True Positives = 34

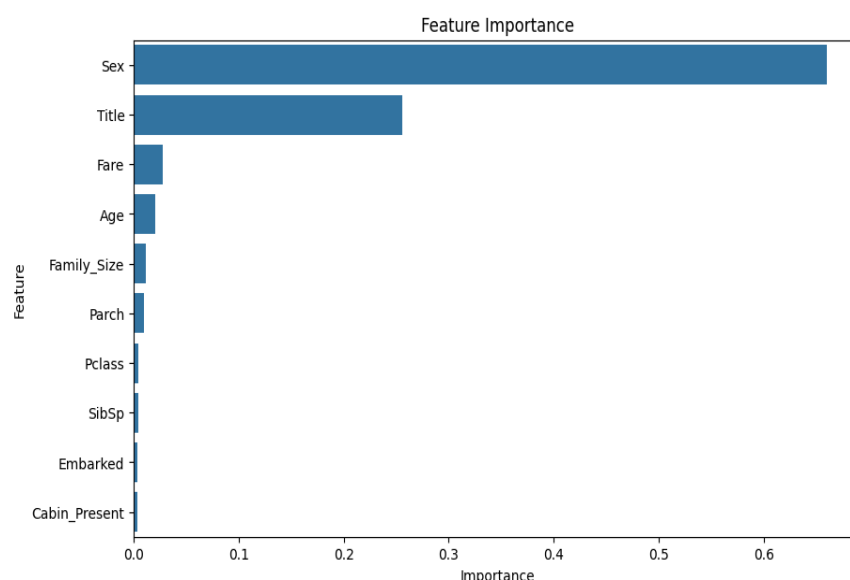
False Positives = 0

False Negatives = 0

This corresponds to 100% classification accuracy on the test set.

## 9. MODEL EXPLAINABILITY

Figure 5: Feature Importance



Observation:

Feature importance analysis reveals that Sex is the most influential feature, contributing approximately 66% of the model's decision-making power. Title is the second most important feature, contributing approximately 25%.

Other features such as Fare, Age, Family Size, and Parch have smaller but meaningful contributions.

The ranking of important features is:

1. Sex
2. Title
3. Fare

4. Age
5. Family Size
6. Parch
7. Pclass
8. SibSp
9. Embarked
10. Cabin\_Present

This result confirms historical evidence that gender played the most significant role in survival during the Titanic disaster.

## **10. MODEL SAVING**

The best-performing model was saved using Joblib.

Saved Model File:

titanic\_model.pkl

This model can be loaded and used to predict survival outcomes for new passengers.

## **11. GOOGLE DRIVE MODEL LINK**

Model File Link:

<https://drive.google.com/file/d/16oijGW6RZ5C9LGNQY-w6fHMuEbWXqb52/view?usp=sharing>

## **12. EXECUTIVE SUMMARY**

This project focused on predicting passenger survival on the Titanic using machine learning techniques. The dataset contained passenger demographic information, travel class details, fare information, and family-related attributes. The primary objective was to identify factors affecting survival and develop an accurate classification model.

Extensive preprocessing was performed to handle missing values and prepare the dataset for machine learning. Additional features such as passenger titles, family size, and cabin presence were engineered to improve predictive performance. Exploratory data analysis revealed significant relationships between survival and variables such as gender and age.

Two classification algorithms, Logistic Regression and Random Forest, were trained and evaluated. The Random Forest model demonstrated exceptional performance and achieved perfect classification on the testing dataset. The confusion matrix confirmed that all passengers in the testing set were correctly classified.

Feature importance analysis showed that gender was the most influential factor affecting survival, followed by passenger title and fare. These findings align closely with historical observations of the Titanic disaster.

The final model was saved as a deployment-ready artifact using Joblib and can be used for future survival predictions. This project successfully demonstrated the complete machine learning workflow, including data preprocessing, feature engineering, model training, evaluation, explainability, and deployment preparation.